## ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** a feather | Air resistance matters for feathers. |
| 2 | **(d)** slowly stops swinging | Friction at the pivot and air resistance. |
| 3 | **(a)** 2500 | 1 g/cm3 = 1000 kg/m3. |
| 4 | **(a)** 8 m/s | 100/12.5. |
| 5 | **(a)** supports the law | Agreement supports it. |
| 6 | **(b)** 20 | 72 x 5/18. |
| 7 | **(c)** 16.7 m/s | 60 km/h x 5/18. |
| 8 | **(d)** low pressure and high temperature | Real gases approach ideal behaviour. |
| 9 | **(c)** 0.15 | 0.5 x 0.3. |
| 10 | **(c)** 0.2 | 2 x 0.5 x 0.2. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Acceptable: air hockey puck on a smooth table (1); Fails: car braking on a rough road (1). | 1+1 |
| 12 | Easy to read and compare (1); 3 x 10^8 (1). | 1+1 |
| 13 | Hypothesis: proposed explanation (1); Prediction: expected result from the hypothesis (1). | 1+1 |
| 14 | Sum = 25.5 (1); Mean = 5.1 g (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | T = 32/20 = 1.6 s (1); Reduces reaction-time error (1); Gives better precision (1). | 1+1+1 |
| 16 | Atom about 10^-10 m (1); Sun about 1.5 x 10^11 m (1); Easy comparison (1). | 1+1+1 |
| 17 | Points follow a trend (1); A stray point stands out (1); Check and repeat (1). | 1+1+1 |
| | **OR:** 0.07 mm = 7 x 10^-5 m (1); = 70 micrometres (1); 1 mm = 1000 micrometres (1). | 1+1+1 |
| 18 | Distance light travels in one year (1); About 9.46 x 10^15 m (1); Used for stars because they are very far (1). | 1+1+1 |
| | **OR:** Wear goggles: protect the eyes (1); Do not taste: poisons (1); Add acid to water slowly: avoid splashes (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Periods: 1.0, 1.42, 1.74, 2.0 s (1); T increases with length (1); Increase is not proportional (1); No friction or air resistance assumed (1); Small amplitude assumed (1). | 1+1+1+1+1 |
| | **OR:** Initial observation (1); Hypothesis (1); Repeated experiments (1); Peer review (1); Wide acceptance (1). | 1+1+1+1+1 |
| 20 | Air breathed in a day (1); Heartbeats in a life (1); Mass of water in a school tank (1); Assumed average values (1); Uncertainty from approximations (1). | 1+1+1+1+1 |
| | **OR:** Mix of litres and gallons or kg and pounds (1); Wrong fuel load (1); Safety risks (1); Need for SI units (1); Double-checking (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Multiplying by $18/5$ instead of $5/18$ (or the reverse) when converting between km/h and m/s; remember that m/s values are always *smaller* numbers.
- Writing a measured value without its unit, or mixing units (cm with m) in one calculation.
- Confusing *law* with *theory*: a law describes a pattern, a theory explains it; a theory does not 'become' a law.
- Taking the mean of the readings as 'the true value'; it is only a more reliable estimate, and a single reading is not 'wrong' just because it differs from others.
- Treating a model as an exact copy of reality, or saying that a model that fails a test should be ignored instead of being improved.
- Mixing up accuracy and precision, and reading the least count of an instrument wrongly (use the smallest division, not the numbered one).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** defines its limits and develops a better model | Model improvement. |
| 2 | **(c)** 12.6 g | Average. |
| 3 | **(a)** 2500 cubic cm | 2.5 x 1000. |
| 4 | **(a)** 'Beauty is the best colour of a flower.' | Cannot be measured. |
| 5 | **(a)** 2 per cent | 1/50 x 100. |
| 6 | **(c)** check the data and refine the model | Honest approach. |
| 7 | **(a)** The liquid has a mass of 30 g. | Has a number. |
| 8 | **(b)** 3 x 10^5 | Divide by 1000. |
| 9 | **(b)** 4 | 1, 2, 4, 3. |
| 10 | **(a)** 5.2 cm | Subtract the zero error. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Error = 0.2 cm (1); 0.2/10 x 100 = 2 per cent (1). | 1+1 |
| 12 | Standard convention in science (1); Prevents confusion (1). | 1+1 |
| 13 | (a) 1.5 m (1); (b) 1500 mm (1). | 1+1 |
| 14 | Smallest value measurable by the instrument (1); 1 mm (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Initial: heavier ball falls faster (1); Experiment: both fall together (1); Modified: fall depends on air resistance not mass (1). | 1+1+1 |
| 16 | Reaction time at start and stop (1); Repeat measurements (1); Average values (1). | 1+1+1 |
| 17 | Newton's laws: motion (1); Law of conservation of mass: reactions (1); Law of conservation of energy: energy changes (1). | 1+1+1 |
| | **OR:** Accuracy: closeness to true value (1); Precision: closeness of repeated readings (1); A balance with zero error is precise but inaccurate (1). | 1+1+1 |
| 18 | Independent: light (1); Dependent: colour (1); Tomatoes ripen in both; heat and chemical changes drive ripening (1). | 1+1+1 |
| | **OR:** Clear communication (1); Same meaning everywhere (1); Avoids errors (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Dalton: indivisible atom (1); Thomson: electrons in a positive sphere (1); Rutherford: nucleus (1); Bohr: shells (1); Each model explained new facts (1). | 1+1+1+1+1 |
| | **OR:** Reports and journals (1); Peer review (1); Honesty prevents false data (1); Humility accepts correction (1); Cooperation across countries like CERN (1). | 1+1+1+1+1 |
| 20 | Accuracy (1); Precision (1); Least count (1); Zero error corrected by subtraction (1); Examples (1). | 1+1+1+1+1 |
| | **OR:** One plot is not enough (1); Need a control plot (1); Same soil and water (1); Repeat on many plots (1); Statistical comparison (1). | 1+1+1+1+1 |

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | Class 9 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** m/s | Combination of base units. |
| 2 | **(b)** an estimate by comparison | Counting units. |
| 3 | **(a)** 1 mm | Smallest division. |
| 4 | **(b)** one-thousandth | 10^-3. |
| 5 | **(d)** 31622400 | 366 x 86400. |
| 6 | **(d)** 0.012 cm | 1.2/100. |
| 7 | **(c)** one-millionth | 10^-6. |
| 8 | **(d)** 3000 | 3 x 1000. |
| 9 | **(c)** air resistance is negligible | Common idealisation. |
| 10 | **(d)** 1000000 | 10^6. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Confirms results are not due to chance (1); Detects hidden errors (1). | 1+1 |
| 12 | Cannot place it on a balance (1); Use gravitational laws and planetary motion (1). | 1+1 |
| 13 | Cannot answer questions that cannot be tested (1); Results depend on measurement accuracy (1). | 1+1 |
| 14 | Beats per year = 72 x 60 x 24 x 365 = 3.8 x 10^7 (1); In 70 years about 2.6 x 10^9 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Easy to picture (1); Electrons do not move in fixed orbits (1); Later models were developed (1). | 1+1+1 |
| 16 | Observation: dark clouds (1); Hypothesis: clouds bring rain (1); Prediction: rain will fall (1). | 1+1+1 |
| 17 | Estimate roughly the expected size (1); Compare the exact calculation (1); A big difference signals an error (1). | 1+1+1 |
| | **OR:** (a) 30 m/s (1); (b) 54 km/h (1); (c) 5000 kg/m3 (1). | 1+1+1 |
| 18 | System is complex (1); Small differences grow (1); Data are limited (1). | 1+1+1 |
| | **OR:** Evidence can be checked (1); Authorities may be wrong (1); Ideas change with data (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Mean = 1.83 mm (1); Error = 0.03 (1); Percentage = 1.67 per cent (1); Readings are precise (1); Slight inaccuracy (1). | 1+1+1+1+1 |
| | **OR:** Number of reliable digits (1); 12.43 s has four (1); Rounding to 12.4 s three figures (1); Final answer cannot be more precise than the data (1); Zeros rules (1). | 1+1+1+1+1 |
| 20 | Observation of falling bodies and planets (1); Mathematical description (1); Predictions tested (1); Agreement supports the law (1); Limits recognised (1). | 1+1+1+1+1 |
| | **OR:** Questions about matter (1); Experiments (1); Atomic models (1); Applications: medicines, batteries (1); Responsibility for the environment (1). | 1+1+1+1+1 |

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